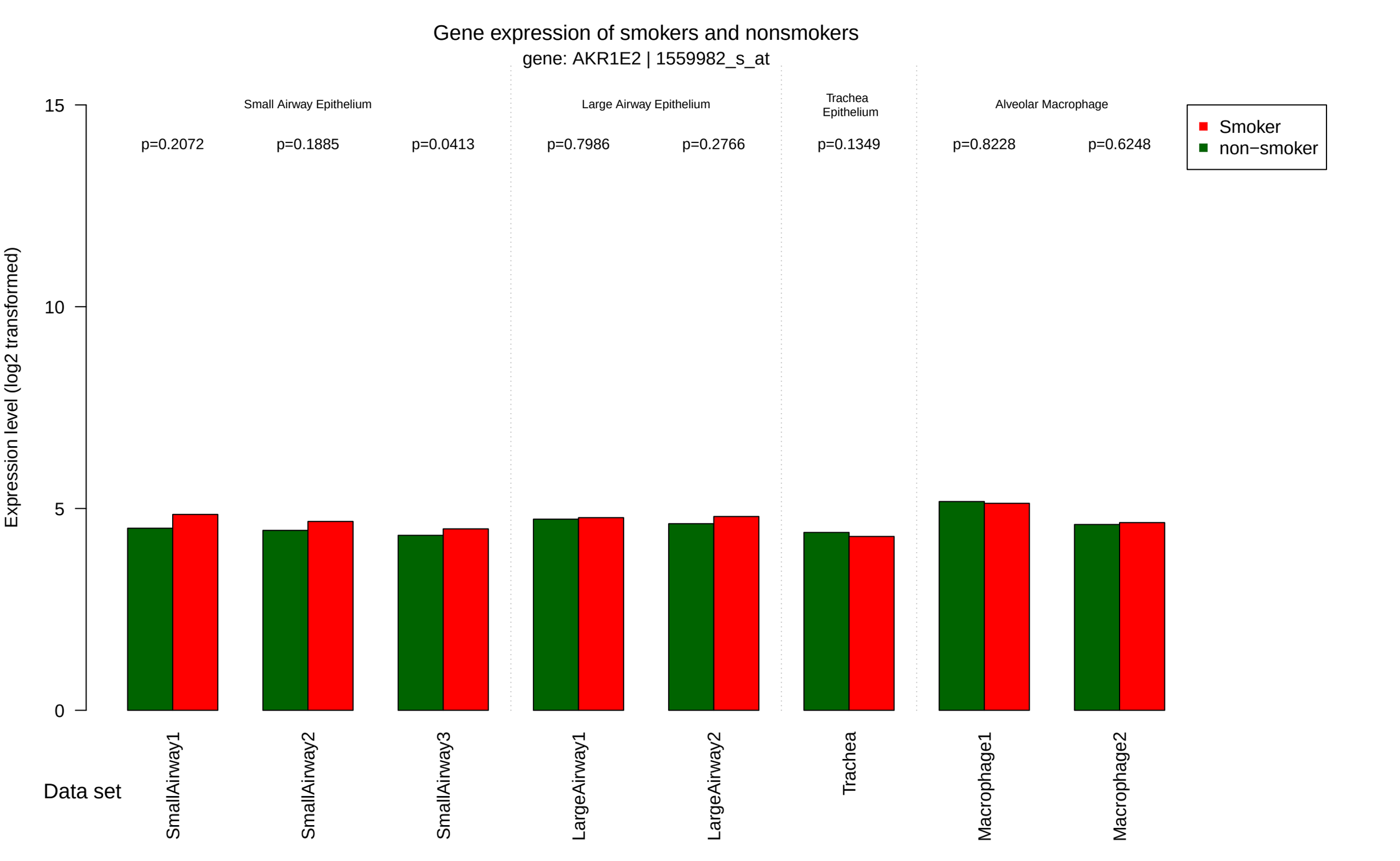
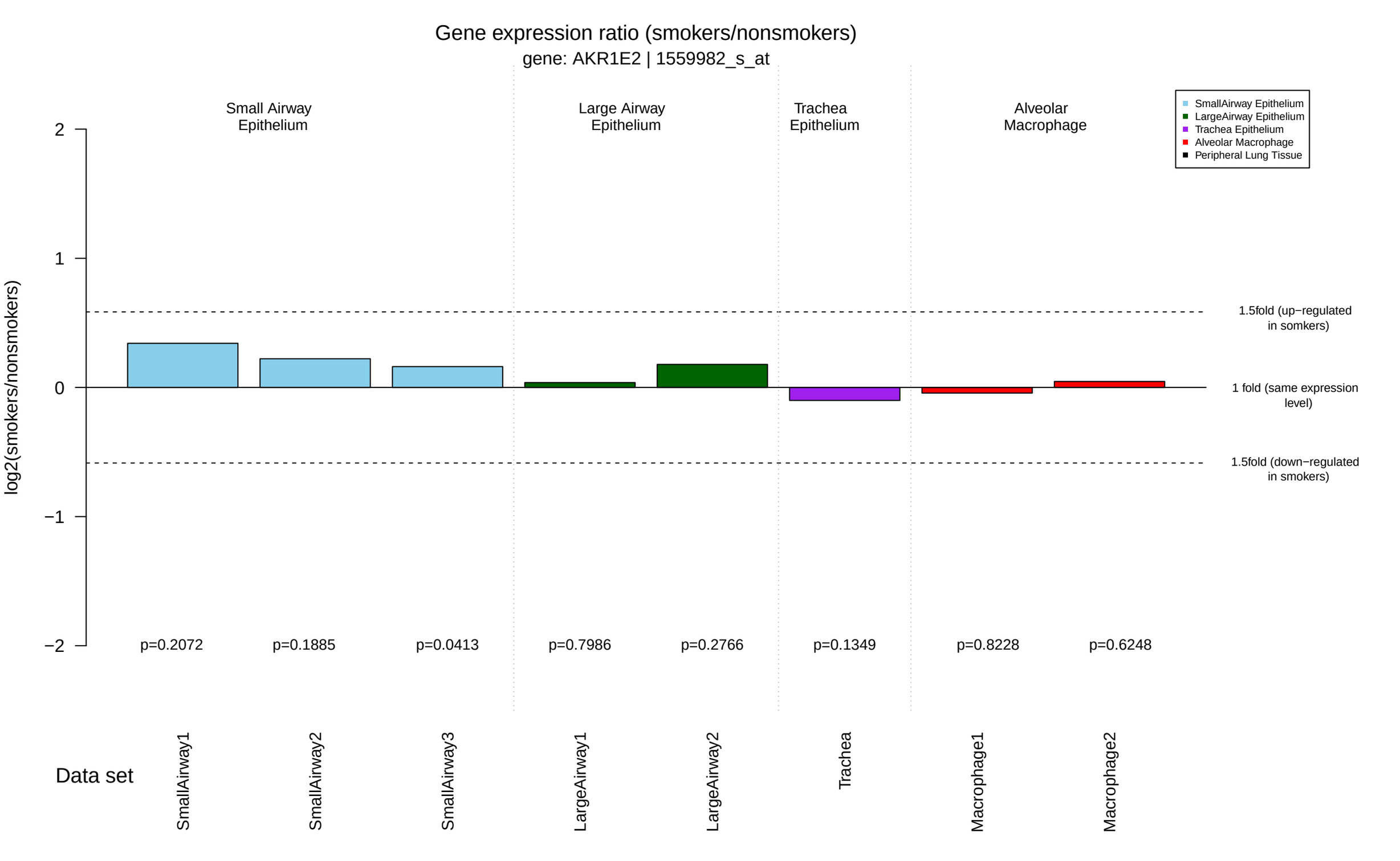
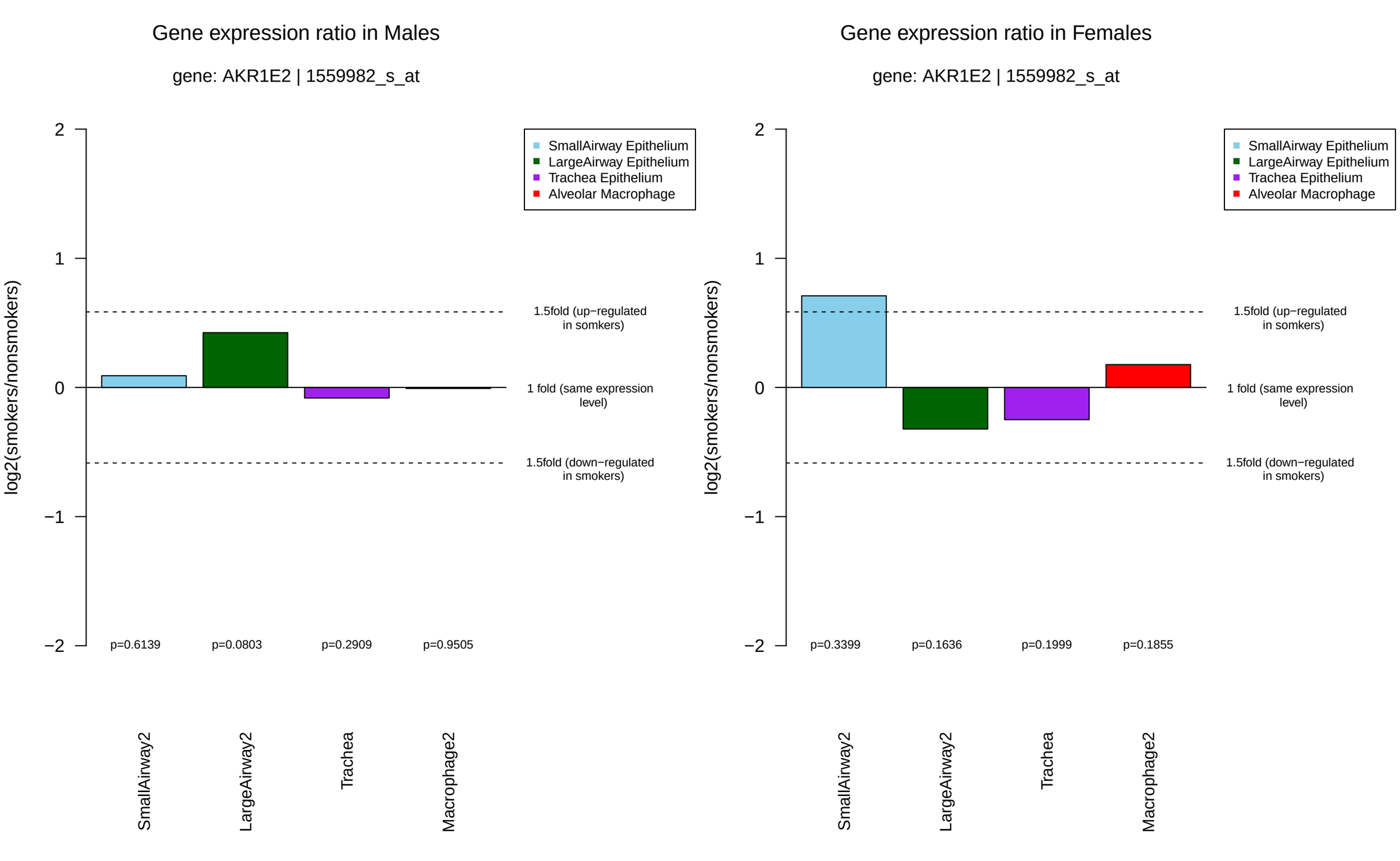
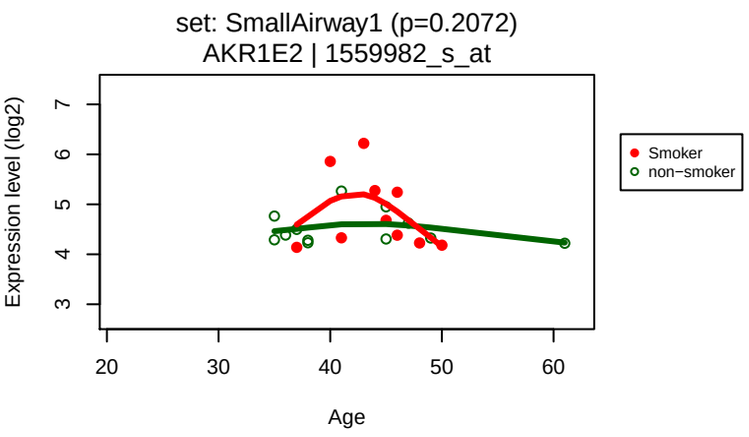
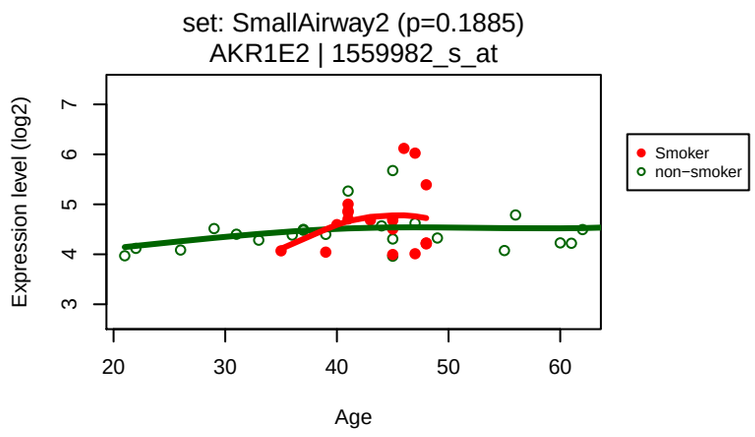
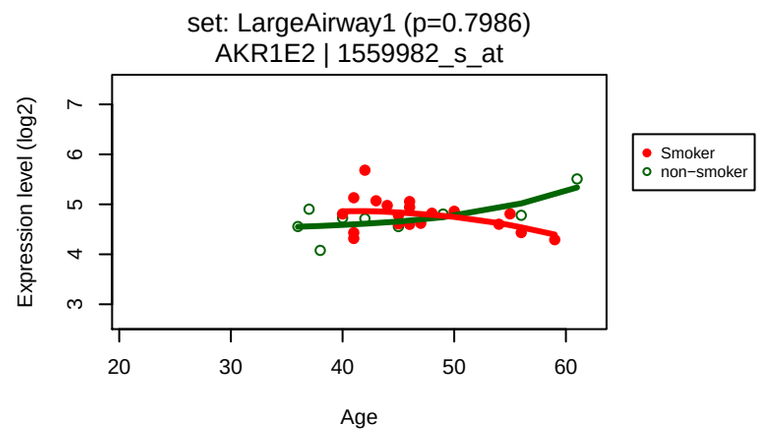

Gene expression of smokers and nonsmokers
gene: AKR1E2 | 1559982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2072
p=0.1885
p=0.0413
p=0.7986
p=0.2766
p=0.1349
p=0.8228
p=0.6248
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

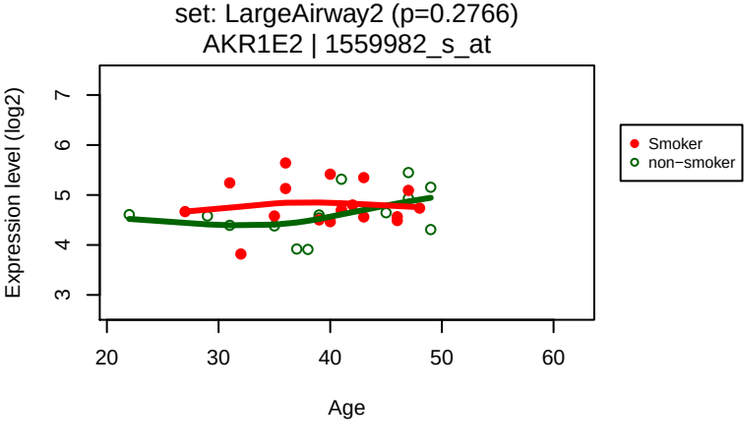

Gene expression ratio (smokers/nonsmokers)
gene: AKR1E2 | 1559982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2072
p=0.1885
p=0.0413
p=0.7986
p=0.2766
p=0.1349
p=0.8228
p=0.6248
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

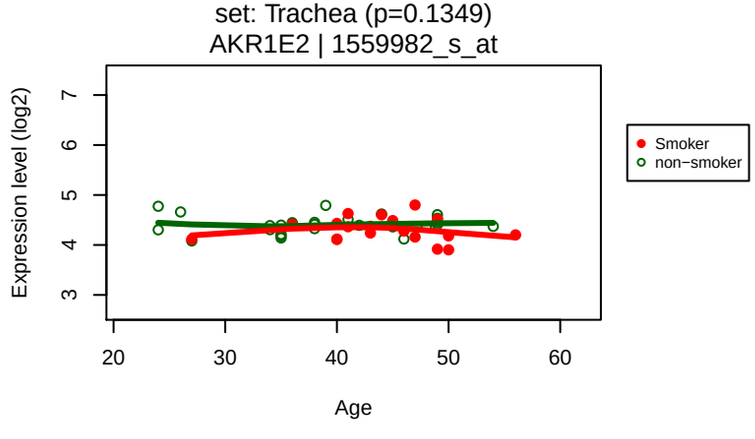

Gene expression ratio in Males
gene: AKR1E2 | 1559982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6139
p=0.0803
p=0.2909
p=0.9505
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1E2 | 1559982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3399
p=0.1636
p=0.1999
p=0.1855
SmallAirway2
LargeAirway2
Trachea
Macrophage2

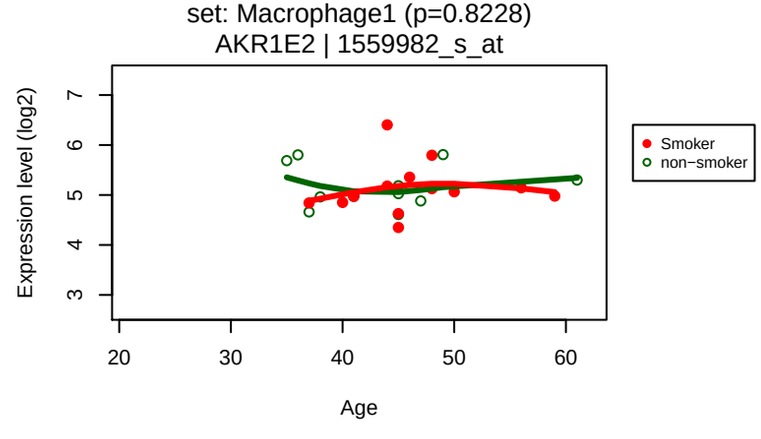

set: SmallAirway1 (p=0.2072)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker

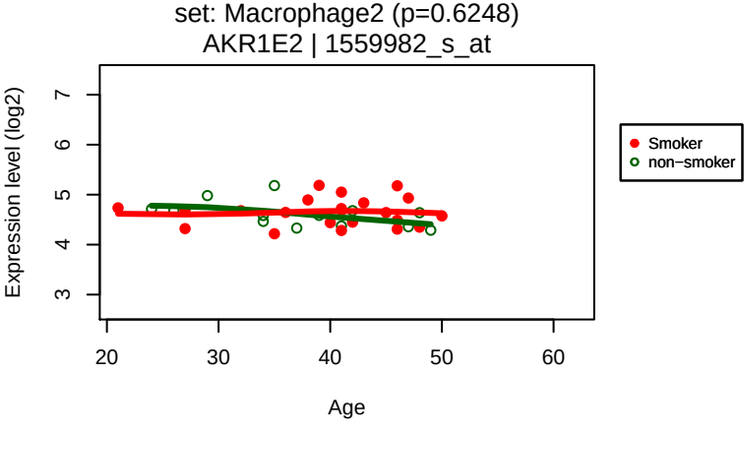

set: SmallAirway2 (p=0.1885)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker

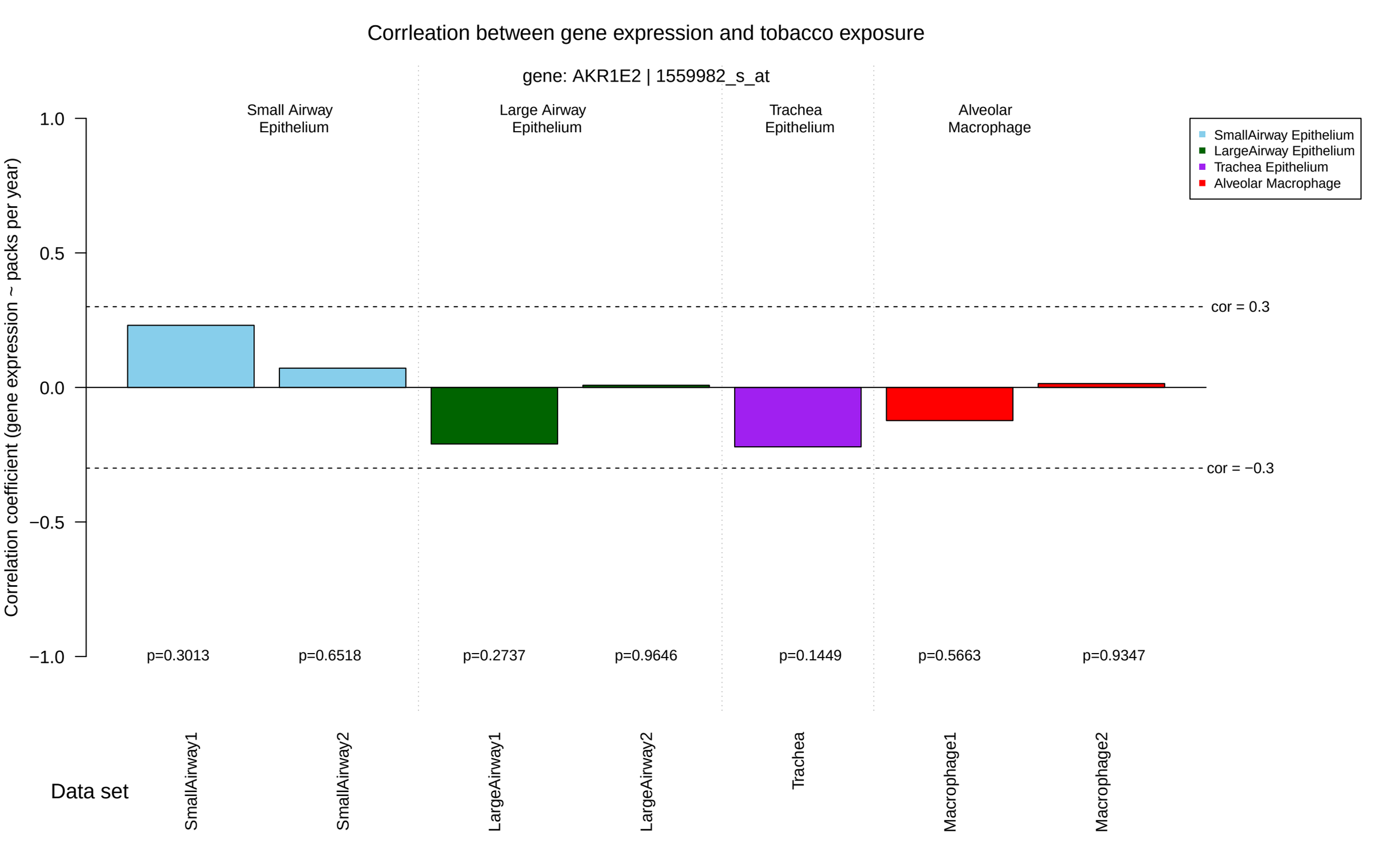

set: LargeAirway1 (p=0.7986)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2766)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1349)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8228)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6248)
AKR1E2 | 1559982_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1E2 | 1559982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3013
p=0.6518
p=0.2737
p=0.9646
p=0.1449
p=0.5663
p=0.9347
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2